

Sparse production radial-Gram decomposition and global owner bound

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-165 left the amplitude annulus $21 < G < 274$ open on one actual R-153 production principal fibre. This note closes that annulus analytically. On the fixed side-16 torus retain one strict-past antipodal pair at p , one at $2p$, and one fresh final pair at $4p$, with unit retained regulator multipliers. Let

$$g = (g_p, g_{2p}) \in \mathbb{R}^{24}, \quad G = |g|, \quad \alpha \in \mathbb{R}^{12}. \quad (1.1)$$

For the exact R-153 endpoint-plus-sixth owner, excluding the explicit source Hessian, the new bound is

$$\boxed{K_{\text{owner}}(g)[\alpha, \alpha] \geq \tilde{P}(G)|\alpha|^2}, \quad (1.2)$$

where

$$\boxed{\tilde{P}(G) = \frac{625}{21040201728}G^4 - \frac{25995}{87359488}G^2 - \frac{1323}{8192000}G - \frac{4402893}{30051663872}}. \quad (1.3)$$

Moreover,

$$\boxed{\tilde{P}(G) > -\frac{9}{10} \quad \text{for every } G \geq 0}. \quad (1.4)$$

Since $10/11 - 1/110 = 9/10$, this proves the R-164 complete-owner target with $\rho = 1/110$ on this entire sparse fibre. It supersedes the R-165 annular subproblem; it does not prove the complete multi-root owner, T-050, A13, a physical phase, a preferred morphology, a PDE, or Sector A.

2. Exact radial-Gram decomposition

Let P be the real orthogonal projector onto the complex doublet. For a six-real field w , set

$$d = |w|^2 + e, \quad \lambda = \frac{|Pw|^2}{d}, \quad b = 2Pw, \quad a = 2(Pw - \alpha_{\text{rad}}\lambda w), \quad \alpha_{\text{rad}} = \frac{5}{9}. \quad (2.1)$$

If B_J is the three-row Pauli-current Gram and B_L the conormal Gram, the R-130 Fierz identity gives the exact matrix equality

$$B_L = B_J + aa^T - bb^T. \quad (2.2)$$

The A1 coefficients are

$$c_0 = \frac{3}{250P_X}, \quad c_1 = \frac{243}{8000P_X}, \quad P_X = 4 + 10^{-12}, \quad (2.3)$$

and hence

$$\boxed{B = (c_0 + c_1)B_J + c_1C, \quad C = aa^T - bb^T}. \quad (2.4)$$

This isolates the only rational radial correction instead of applying the R-130 absolute derivative envelope to all six rows.

3. Exact Pauli resonance cancellation

Condition on the strict past and write

$$W = m + X, \quad n = \partial m, \quad z = S_{4p}\alpha, \quad u = \partial z. \quad (3.1)$$

For every real symmetric embedded Pauli generator S_A , put

$$J_A(t) = 2(W + tz)^T S_A(n + tu). \quad (3.2)$$

Then

$$J_A''(0) = 4z^T S_A u = 2\partial(z^T S_A z). \quad (3.3)$$

After the centered fresh expectation, $\mathbb{E}J_A(0) = 2m^T S_A n$ has phase frequencies of magnitude at most four. Equation (3.3) has only frequencies ± 8 : its zero coefficient vanishes because it is a periodic derivative. Consequently

$$\int \mathbb{E}[J_A(0)J_A''(0)] dx = 0. \quad (3.4)$$

The remaining current contribution is $\int \mathbb{E}|J_A'(0)|^2 dx \geq 0$. The unaveraged Xn term also cannot resonate: its frequencies have magnitude two, three, five, or six. The strict-past trace debit is not discarded; it is paid separately in Section 6. Thus (3.4) is a current cancellation, not a claim that the complete Pauli owner is positive.

4. Floor-uniform radial correction bounds

Put $R = (P - \lambda I)w$. Direct differentiation gives

$$\nabla \lambda = \frac{2R}{d}, \quad D^2 \lambda = \frac{2(P - \lambda I)}{d} - \frac{4(wR^T + Rw^T)}{d^2}. \quad (4.1)$$

Writing $x = |Pw|^2/d$ and $\delta = e/d$ shows

$$\frac{|R|^2}{d} = x(1 - x) - x^2\delta \leq x(1 - x) \leq \frac{1}{4}, \quad \frac{|w|^2}{d} \leq 1. \quad (4.2)$$

Therefore

$$|w||\nabla \lambda| \leq 1, \quad |w|^2 \|D^2 \lambda\|_{\text{op}} \leq 6. \quad (4.3)$$

The apparent division by $|w|$ below is removable at $w = 0$. Equations (2.1) and (4.3) imply

$$|a|, |b| \leq 2|w|, \quad \|Da\| \leq \frac{38}{9}, \quad \|Db\| \leq 2, \quad |D^2 a[z, z]| \leq \frac{80}{9} \frac{|z|^2}{|w|}. \quad (4.4)$$

For every real symmetric Q this yields

$$|C : Q| \leq 8\|Q\|_{\text{op}}|w|^2, \quad (4.5)$$

$$|D(C : Q)[z]| \leq \frac{224}{9}\|Q\|_{\text{op}}|w||z|, \quad (4.6)$$

$$\left| \frac{1}{2} D^2(C : Q)[z, z] \right| \leq \frac{3208}{81}\|Q\|_{\text{op}}|z|^2. \quad (4.7)$$

Indeed the three terms in (4.7) are $1444/81$, $1440/81$, and $324/81$. Since $c_0 < 3/1000$ and $c_1 < 243/32000$, the outward rational constants are

$$L_C = c_1 \frac{224}{9} < \frac{189}{1000}, \quad H_C = c_1 \frac{3208}{81} < \frac{1203}{4000}, \quad (4.8)$$

$$H_{\text{tr}} = 12(c_0 + c_1) + H_C < \frac{3423}{8000}. \quad (4.9)$$

5. Direct harmonic coercivity

The R-165 factor $5/6$ is sharp relative to $\langle |m|^4 \rangle \langle |z|^2 \rangle$, but that product is not the quantity needed for the source-coordinate lower bound. A stronger direct coefficient estimate is available. Put $r = |m|^2$, $f = r^2$, and $q = |z|^2$. Then q has only modes $0, \pm 8$, $f_8 = r_4^2$, and positivity gives $q_0 \geq 2|q_8|$. Parseval gives

$$f_0 - |f_8| = r_0^2 + 2 \sum_{k=1}^3 |r_k|^2 + |r_4|^2 \geq r_0^2. \quad (5.1)$$

Hence

$$\boxed{\langle |m|^4 |z|^2 \rangle \geq \langle |m|^2 \rangle^2 \langle |z|^2 \rangle.} \quad (5.2)$$

This does not contradict the sharp $5/6$ result, because $\langle |m|^4 \rangle \geq \langle |m|^2 \rangle^2$ can have strict slack. Centered Gaussian fourth-moment expansion only increases the retained term, so the sixth Hessian supplies

$$A = \frac{9}{10} \frac{(100/63)^2 (20/91)}{4096^2} = \frac{625}{21040201728}. \quad (5.3)$$

The other sixth-Hessian square is left unused.

6. One-use coefficient ledger

Use the exact R-165 covariance bounds

$$\frac{100}{63} \mathbf{I} \leq \Gamma_p, \Gamma_{2p} \leq \frac{100}{31} \mathbf{I}, \quad \frac{20}{91} \mathbf{I} \leq \Gamma_{4p} \leq \frac{10}{43} \mathbf{I}, \quad \text{Tr } \Gamma_{4p} \leq \frac{60}{43}. \quad (6.1)$$

Also $p^2 < 5/32$, the past derivative covariance is at most $625/507904$, and $\sqrt{2(100/31)(60/43)} < 301/100$. The correction value term, its first derivative, its half-Hessian, the fresh value covariance, and the complete strict-past trace are each paid once. They give

$$\begin{aligned} B_2 &= \frac{25995}{87359488}, & B_1 &= \frac{1323}{8192000}, \\ D_{\text{fresh}} &= \frac{729}{30294016}, & D_{\text{trace}} &= \frac{85575}{698875904}, \end{aligned} \quad (6.2)$$

and

$$D_0 = D_{\text{fresh}} + D_{\text{trace}} = \frac{4402893}{30051663872}. \quad (6.3)$$

The raw dimensionless coefficient in B_2 is

$$256 + 32 \frac{\cdot 224}{9} + 8 \frac{\cdot 3208}{81}; \quad (6.4)$$

the three terms are respectively the correction value, first derivative, and half-Hessian. The Pauli current square is not counted as a reserve, and the source $9\mathbf{I}/10$ remains outside K_{owner} .

7. Exact global minimum certificate

For $\tilde{P}$ in (1.3), exact arithmetic gives

$$\tilde{P}'(70) = -\frac{1360786403}{1277632512000} < 0, \quad \tilde{P}'(71) = \frac{97738714417}{876455903232000} > 0, \quad (7.1)$$

and

$$\tilde{P}''(70) = \frac{6865745}{5962285056} > 0. \quad (7.2)$$

Since $\tilde{P}''$ has exactly one positive zero and is increasing after it, $\tilde{P}'$ has exactly one nonnegative zero G_* , with $70 < G_* < 71$. Convexity on that interval gives the rational lower bound

$$\begin{aligned}\tilde{P}(G_*) &\geq \tilde{P}(70) + \tilde{P}'(70) \\ &= -\frac{332863942666997}{439505584128000}.\end{aligned}\tag{7.3}$$

Its margin above the target is

$$\frac{9}{10} + \tilde{P}(G_*) \geq \frac{62691083048203}{439505584128000} > 0.\tag{7.4}$$

This proves (1.4) on the full radial half-line and closes the R-165 annulus without numerical quadrature, interval subdivision, or an exact Gaussian matrix evaluation.

8. Route decision and Devil's-advocate review

The useful lesson is methodological. A direct high-dimensional interval attack on the rational Gaussian 12×12 matrix was a valid fallback, but it hid the exact Pauli resonance cancellation and the one-row radial defect. That brute-force route is now parked for this fibre. It remains a possible falsifier for future fibres lacking the same support algebra.

1. **Objection: the new harmonic coefficient contradicts the sharp 5/6 fixture. DISMISSED.** The two inequalities have different right-hand sides. Equation (5.2) is proved coefficient by coefficient and is weaker than replacing its right side by $\langle |m|^4 \rangle \langle |z|^2 \rangle$.
2. **Objection: the Pauli trace was silently discarded. DISMISSED.** Only the current resonance vanishes. The full strict-past trace is included in H_{tr} and D_{trace} .
3. **Objection: the positive floor makes (4.3) singular at the origin. DISMISSED.** The bounds follow with $e/d \geq 0$, and C and its displayed products extend continuously at $w = 0$.
4. **Objection: a finite floating search proves the result. UPHELD against that inference.** No sampled value enters (1.3) or (7.4). The certificate is rational and analytic.
5. **Objection: this closes T-050. UPHELD against that inference.** Only one sparse principal fibre is closed. Other past harmonics, final roots, cross-root blocks, low variables, random/nonlinear/revisit feedback, and removal remain open.
6. **Objection: success selects BCC or validates the candidate PDE. UPHELD against that inference.** No morphology is assumed. This is a scoped mathematical test of one candidate action, not a physical-vacuum result.

9. Reproduction and Result footer

Result ID: A13-CLASSII-SPARSE-PRODUCTION-OWNER-RADIAL-GRAM-
GLOBAL-BOUNDARY (R-166).

Precise statement: on the fixed exact p:2p past / fresh-4p conditional fibre,
 $K_{\text{owner}}[g] \geq P_{\text{tilde}}(|g|)$ I with (1.3), and $K_{\text{owner}} > -9I/10$ for all g.

Dependencies: A1; R-130, R-153, R-164, R-165.

Scope: fixed side-16 finite chart, positive floor, exact continuum torus
average, unit retained multipliers, only the declared p:2p past and 4p root.

Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.

Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_sparse_production_owner_radial_gram_
global_boundary.py

Expected output: primary 34/34, independent 31/31, integrated verification,
deterministic five-page PDF, and all release gates pass.

Independent command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_sparse_production_owner_radial_gram_
global_boundary_independent.py

Integrated command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_sparse_production_owner_radial_gram_
global_boundary_verify.py

Falsification gate: failure of the Gram split, Pauli resonance cancellation,
radial derivative bounds, direct harmonic lemma, one-use factor ledger,
global rational minimum, authority hashes, PDF, records, or scope firewall.

Tier before / after: T4 / T4; no promotion.

Proof complete: false. T-050 closed: false. Sector A closed: false.

No-overclaim statement: no complete multi-root owner, feedback/removal,
T-050, A13, Nelson/measure, phase, morphology, PDE, or Sector A theorem.

Next required action: test and assemble the nonzero complete multi-root owner;
keep T-054/T-052 truth-first candidate evaluation active in parallel.